

Composable Deep Reinforcement Learning for Robotic Manipulation

Item Type

Conference Paper

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Abstract

Model-free deep reinforcement learning has been shown to exhibit good performance in domains ranging from video games to simulated robotic manipulation and locomotion. However, model-free methods are known to perform poorly when the interaction time with the environment is limited, as is the case for most real-world robotic tasks. In this paper, we study how maximum entropy policies trained using soft Q-learning can be applied to real-world robotic manipulation. The application of this method to real-world manipulation is facilitated by two important features of soft Q-learning. First, soft Q-learning can learn multimodal exploration strategies by learning policies represented by expressive energy-based models. Second, we show that policies learned with soft Q-learning can be composed to create new policies, and that the optimality of the resulting policy can be bounded in terms of the divergence between the composed policies. This compositionality provides an especially valuable tool for real-world manipulation, where constructing new policies by composing existing skills can provide a large gain in efficiency over training from scratch. Our experimental evaluation demonstrates that soft Q-learning is substantially more sample efficient than prior model-free deep reinforcement learning methods, and that compositionality can be performed for both simulated and real-world tasks.

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2018-05

Library Catalog

IEEE Xplore

URL

https://ieeexplore.ieee.org/abstract/document/8460756?casa_token=2kTmnF72JdUAAAAA:YHhJJQj2q07OMCheJv8TeanimTdqv1qQGqHRv8Cj__Nh2MWBQD7-VvDdZVHXoqrElounGE2kaA

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2018 IEEE International Conference on Robotics and Automation (ICRA)

Conference Name

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Tags:

Task analysis, Entropy, Learning (artificial intelligence), Machine learning, Neural networks, Robots, Training

Attachments

- IEEE Xplore Abstract Record

Composable energy policies for reactive motion generation and reinforcement learning

Item Type

Journal Article

Author

Julen Urain

Author

Anqi Li

Author

Puze Liu

Author Carlo D'Eramo
Author Jan Peters
Abstract In this work, we introduce composable energy policies (CEP), a novel framework for multi-objective motion generation. We frame the problem of composing multiple policy components from a probabilist...
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Language en
Loc. in Archive Sage UK: London, England
Library Catalog journals.sagepub.com
URL <https://journals.sagepub.com/doi/epub/10.1177/02783649231179499>
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Extra Publisher: SAGE PublicationsSage UK: London, England
Publication The International Journal of Robotics Research
DOI 10.1177/02783649231179499
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Attachments

- Snapshot

Composite Motion Learning with Task Control

Item Type Journal Article
Author Pei Xu
Author Xiumin Shang
Author Victor Zordan
Author Ioannis Karamouzas
Abstract We present a deep learning method for composite and task-driven motion control for physically simulated characters. In contrast to existing data-driven approaches using reinforcement learning that imitate full-body motions, we learn decoupled motions for specific body parts from multiple reference motions simultaneously and directly by leveraging the use of multiple discriminators in a GAN-like setup. In this process, there is no need of any manual work to produce composite reference motions for learning. Instead, the control policy explores by itself how the composite motions can be combined automatically. We further account for multiple task-specific rewards and train a single, multi-objective control policy. To this end, we propose a novel framework for multi-objective learning that adaptively balances the learning of disparate motions from multiple sources and multiple goal-directed control objectives. In addition, as composite motions are typically augmentations of simpler behaviors, we introduce a sample-efficient method for training composite control policies in an incremental manner, where we reuse a pre-trained policy as the meta policy and train a cooperative policy that adapts the meta one for new composite tasks. We show the applicability of our approach on a variety of challenging multi-objective tasks involving both composite motion imitation and multiple goal-directed control.
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Library Catalog arXiv.org
URL <http://arxiv.org/abs/2305.03286>
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Extra arXiv:2305.03286 [cs]
Volume 42
Pages 1-16
Publication ACM Transactions on Graphics
DOI 10.1145/3592447

Citation Key xuCompositeMotionLearning2023
Issue 4
Journal Abbr ACM Trans. Graph.
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Tags:

Computer Science - Machine Learning, Computer Science - Artificial Intelligence, Computer Science - Graphics

Notes:

Comment: SIGGRAPH 2023. Code: <https://github.com/xupei0610/CompositeMotion>. Video: <https://youtu.be/mcRAxwoTh3E>

Attachments

- [arXiv.org Snapshot](#)

Hierarchical Policy Blending as Inference for Reactive Robot Control

Item Type Preprint
Author Kay Hansel
Author Julen Urain
Author Jan Peters
Author Georgia Chalvatzaki
Abstract Motion generation in cluttered, dense, and dynamic environments is a central topic in robotics, rendered as a multi-objective decision-making problem. Current approaches trade-off between safety and performance. On the one hand, reactive policies guarantee fast response to environmental changes at the risk of suboptimal behavior. On the other hand, planning-based motion generation provides feasible trajectories, but the high computational cost may limit the control frequency and thus safety. To combine the benefits of reactive policies and planning, we propose a hierarchical motion generation method. Moreover, we adopt probabilistic inference methods to formalize the hierarchical model and stochastic optimization. We realize this approach as a weighted product of stochastic, reactive expert policies, where planning is used to adaptively compute the optimal weights over the task horizon. This stochastic optimization avoids local optima and proposes feasible reactive plans that find paths in cluttered and dense environments. Our extensive experimental study in planar navigation and 6DoF manipulation shows that our proposed hierarchical motion generation method outperforms both myopic reactive controllers and online re-planning methods.
Date 2024-07-29
Library Catalog arXiv.org
URL <http://arxiv.org/abs/2210.07890>
Accessed 10/24/2024, 5:23:09 PM
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DOI 10.48550/arXiv.2210.07890
Repository arXiv
Citation Key hanselHierarchicalPolicyBlending2024
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Computer Science - Machine Learning, Computer Science - Robotics

Notes:

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Attachments

- Preprint PDF
- Preprint PDF
- Snapshot
- Snapshot

Learning an Embedding Space for Transferable Robot Skills

Item Type	Conference Paper
Author	Karol Hausman
Author	Jost Tobias Springenberg
Author	Ziyu Wang
Author	Nicolas Heess
Author	Martin Riedmiller
Abstract	We present a method for reinforcement learning of closely related skills that are parameterized via a skill embedding space. We learn such skills by taking advantage of latent variables and exploiting a connection between reinforcement learning and variational inference. The main contribution of our work is an entropy-regularized policy gradient formulation for hierarchical policies, and an associated, data-efficient and robust off-policy gradient algorithm based on stochastic value gradients. We demonstrate the effectiveness of our method on several simulated robotic manipulation tasks. We find that our method allows for discovery of multiple solutions and is capable of learning the minimum number of distinct skills that are necessary to solve a given set of tasks. In addition, our results indicate that the hereby proposed technique can interpolate and/or sequence previously learned skills in order to accomplish more complex tasks, even in the presence of sparse rewards.
Date	2018/02/15
Language	en
Library Catalog	openreview.net
URL	https://openreview.net/forum?id=rk07ZXZRb
Accessed	4/8/2024, 11:47:09 AM
Conference Name	International Conference on Learning Representations
Citation Key	hausmanLearningEmbeddingSpace2018
Date Added	4/8/2024, 11:47:09 AM
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Learning from demonstration using products of experts: applications to manipulation and task prioritization

Item Type	Preprint
Author	Emmanuel Pignat
Author	João Silvério
Author	Sylvain Calinon

Abstract Probability distributions are key components of many learning from demonstration (LfD) approaches. While the configuration of a manipulator is defined by its joint angles, poses are often best explained within several task spaces. In many approaches, distributions within relevant task spaces are learned independently and only combined at the control level. This simplification implies several problems that are addressed in this work. We show that the fusion of models in different task spaces can be expressed as a product of experts (PoE), where the probabilities of the models are multiplied and renormalized so that it becomes a proper distribution of joint angles. Multiple experiments are presented to show that learning the different models jointly in the PoE framework significantly improves the quality of the model. The proposed approach particularly stands out when the robot has to learn competitive or hierarchical objectives. Training the model jointly usually relies on contrastive divergence, which requires costly approximations that can affect performance. We propose an alternative strategy using variational inference and mixture model approximations. In particular, we show that the proposed approach can be extended to PoE with a nullspace structure (PoENS), where the model is able to recover tasks that are masked by the resolution of higher-level objectives.

Date 2020-10-07

Short Title Learning from demonstration using products of experts

Library Catalog arXiv.org

URL <http://arxiv.org/abs/2010.03505>

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DOI 10.48550/arXiv.2010.03505

Repository arXiv

Citation Key pignatLearningDemonstrationUsing2020

Archive ID arXiv:2010.03505

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Tags:

Computer Science - Machine Learning, Computer Science - Robotics

Attachments

- Snapshot

Learning to Compose Hierarchical Object-Centric Controllers for Robotic Manipulation

Item Type Preprint

Author Mohit Sharma

Author Jacky Liang

Author Jialiang Zhao

Author Alex LaGrassa

Author Oliver Kroemer

Abstract Manipulation tasks can often be decomposed into multiple subtasks performed in parallel, e.g., sliding an object to a goal pose while maintaining contact with a table. Individual subtasks can be achieved by task-axis controllers defined relative to the objects being manipulated, and a set of object-centric controllers can be combined in an hierarchy. In prior works, such combinations are defined manually or learned from demonstrations. By contrast, we propose using reinforcement learning to dynamically compose hierarchical object-centric controllers for manipulation tasks. Experiments in both simulation and real world show how the proposed approach leads to improved sample efficiency, zero-shot generalization to novel test environments, and simulation-to-reality transfer without fine-tuning.

Date 2020-11-13

Library Catalog arXiv.org

URL <http://arxiv.org/abs/2011.04627>

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Extra arXiv:2011.04627 [cs]
DOI 10.48550/arXiv.2011.04627
Repository arXiv
Citation Key sharmaLearningComposeHierarchical2020
Archive ID arXiv:2011.04627
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Computer Science - Machine Learning, Computer Science - Robotics, Computer Science - Artificial Intelligence

Notes:

Comment: Accepted as Plenary Talk at CoRL'20. First two authors contributed equally. For results see <https://sites.google.com/view/compositional-object-control/>

Attachments

- arXiv.org Snapshot

Manipulation task primitives for composing robot skills

Item Type Conference Paper
Author J.D. Morrow
Author P.K. Khosla
Abstract A manipulation task primitive is classified by the relative motion between two (rigid) parts. Only twenty different relative motions are possible and these can be used to guide the identification and development of manipulation task primitives. The goal is to build a richer library of robot capabilities in the manipulation domain. By identifying manipulation task primitives and instantiating solutions to them with available sensors and robot hardware in the form of sensorimotor primitives, we provide a higher-level abstraction for composing solutions to complex manipulation tasks. A key benefit is the ability to re-use costly sensor-based control algorithms for executing these primitives. We discuss the implementation of a few manipulation task primitives using force damping control and active vision feedback. Finally, we decompose a common task into two different skills using the primitives described.
Date 1997-04
Library Catalog IEEE Xplore
URL https://ieeexplore.ieee.org/abstract/document/606800?casa_token=vtSVk4MC1T8AAAAA:EkHZUB-OxwGN5_oHfUpp6AKSV7RzGPNm-B8ERUHVi0l0LNoN6DMuSzwxxkhOVkmDvAgblIl06UQ
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Volume 4
Pages 3354-3359 vol.4
Proceedings Title Proceedings of International Conference on Robotics and Automation
Conference Name Proceedings of International Conference on Robotics and Automation
DOI 10.1109/ROBOT.1997.606800
Citation Key morrowManipulationTaskPrimitives1997
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Tags:

Robot sensing systems, Force control, Force feedback, Force sensors, Hardware, Libraries, Orbital robotics, Robot control, Robot programming, Robustness

MCP: Learning Composable Hierarchical Control with Multiplicative Compositional Policies

Item Type Preprint
Author Xue Bin Peng
Author Michael Chang
Author Grace Zhang
Author Pieter Abbeel
Author Sergey Levine
Abstract Humans are able to perform a myriad of sophisticated tasks by drawing upon skills acquired through prior experience. For autonomous agents to have this capability, they must be able to extract reusable skills from past experience that can be recombined in new ways for subsequent tasks. Furthermore, when controlling complex high-dimensional morphologies, such as humanoid bodies, tasks often require coordination of multiple skills simultaneously. Learning discrete primitives for every combination of skills quickly becomes prohibitive. Composable primitives that can be recombined to create a large variety of behaviors can be more suitable for modeling this combinatorial explosion. In this work, we propose multiplicative compositional policies (MCP), a method for learning reusable motor skills that can be composed to produce a range of complex behaviors. Our method factorizes an agent's skills into a collection of primitives, where multiple primitives can be activated simultaneously via multiplicative composition. This flexibility allows the primitives to be transferred and recombined to elicit new behaviors as necessary for novel tasks. We demonstrate that MCP is able to extract composable skills for highly complex simulated characters from pre-training tasks, such as motion imitation, and then reuse these skills to solve challenging continuous control tasks, such as dribbling a soccer ball to a goal, and picking up an object and transporting it to a target location.
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Library Catalog arXiv.org
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Accessed 4/1/2024, 2:07:54 PM
Extra arXiv:1905.09808 [cs, stat]
DOI 10.48550/arXiv.1905.09808
Repository arXiv
Citation Key pengMCPLearningComposable2019
Archive ID arXiv:1905.09808
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Tags:
Computer Science - Machine Learning, Statistics - Machine Learning

Attachments
◦ arXiv.org Snapshot

PoCo: Policy Composition from and for Heterogeneous Robot Learning

Item Type Preprint
Author Lirui Wang
Author Jialiang Zhao

Author Yilun Du

Author Edward H. Adelson

Author Russ Tedrake

Abstract Training general robotic policies from heterogeneous data for different tasks is a significant challenge. Existing robotic datasets vary in different modalities such as color, depth, tactile, and proprioceptive information, and collected in different domains such as simulation, real robots, and human videos. Current methods usually collect and pool all data from one domain to train a single policy to handle such heterogeneity in tasks and domains, which is prohibitively expensive and difficult. In this work, we present a flexible approach, dubbed Policy Composition, to combine information across such diverse modalities and domains for learning scene-level and task-level generalized manipulation skills, by composing different data distributions represented with diffusion models. Our method can use task-level composition for multi-task manipulation and be composed with analytic cost functions to adapt policy behaviors at inference time. We train our method on simulation, human, and real robot data and evaluate in tool-use tasks. The composed policy achieves robust and dexterous performance under varying scenes and tasks and outperforms baselines from a single data source in both simulation and real-world experiments.

Date 2024-02-04

Language en

Short Title PoCo

Library Catalog arXiv.org

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Repository arXiv

Citation Key wangPoCoPolicyComposition2024

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Zero-Shot Skill Composition and Simulation-to-Real Transfer by Learning Task Representations

Item Type Journal Article

Author Zhanpeng He

Author Ryan Julian

Author Eric Heiden

Author Hejia Zhang

Author Stefan Schaal

Author Joseph J Lim

Author Gaurav Sukhatme

Author Karol Hausman

Abstract Simulation-to-real transfer is an important strategy for making reinforcement learning practical with real robots. Successful sim-to-real transfer systems have difficulty producing policies which generalize across tasks, despite training for thousands of hours equivalent real robot time. To address this shortcoming, we present a novel approach to efficiently learning new robotic skills directly on a real robot, based on model-predictive control (MPC) and an algorithm for learning task representations. In short, we show how to reuse the simulation from the pre-training step of sim-to-real methods as a tool for foresight, allowing the sim-to-real policy adapt to unseen tasks. Rather than end-to-end learning policies for single tasks and attempting to transfer them, we first use simulation to simultaneously learn (1) a continuous parameterization (i.e. a skill embedding or latent) of task-appropriate primitive skills, and (2) a single policy for these skills which is conditioned on this representation. We then directly transfer our multitask policy to a real robot, and actuate the robot by choosing sequences of skill latents which actuate the policy, with each latent corresponding to a

pre-learned primitive skill controller. We complete unseen tasks by choosing new sequences of skill latents to control the robot using MPC, where our MPC model is composed of the pre-trained skill policy executed in the simulation environment, run in parallel with the real robot. We discuss the background and principles of our method, detail its practical implementation, and evaluate its performance by using our method to train a real Sawyer Robot to achieve motion tasks such as drawing and block pushing.

Language en

Library Catalog Zotero

Citation Key heZeroShotSkillComposition

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